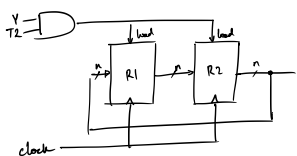
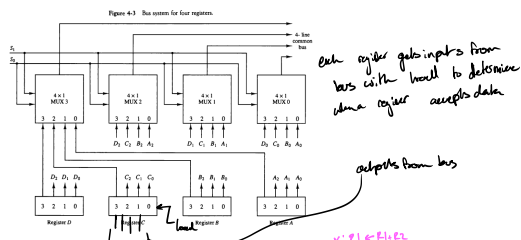


1 4 5 8 11 14 16 18 20 25

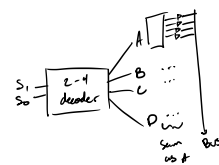
4-1 Block diagram for $yT_2 \leftarrow R2 \leftarrow R1, R1 \leftarrow R2$



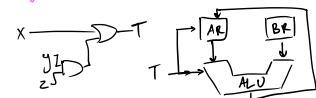
4-4 What has to be done to the bus to transfer information from register to register. Show the connections that must be included to provide a path for each reg. C4 the inputs of register A.



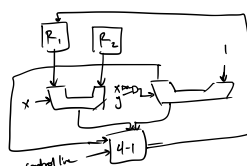
4-5 Pass a Bus with 3-bit buses & a decoder instead of a multiplexer



4-8 Draw the block diagram for the hardware to implement: $x \leftarrow yz$; $AR \leftarrow AR + B$



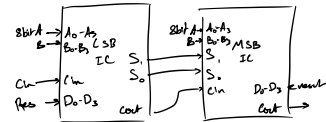
4-11 Draw a block diagram to show the following: $x \leftarrow R1 + R2$; $y \leftarrow R1 + R1 + 1$



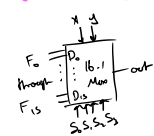
4-12 determine values of the outputs: S_3, S_2, S_1, S_0, C_4

S#	M	A	B	$S_3 S_2 S_1 S_0$	C_4
1	1	0111	0110	0000	1
2	1	1000	1001	1110	0
3	0	1100	1000	0100	1
4	0	0101	1010	1111	0
5	0	0000	0001	0001	0

4-14 Show connections among two ICs to form an 8 bit arithmetic circuit



4-16 Derive a combinational circuit that selects 3 genes out of the 16 logic functions listed in T. 4-5



4-18 Register A holds the 8-bit binary 11011001. Find B op's logic micro ops to be performed in order to change the value A to: a) 0110101 b) 11111101

a) B op : 01111111
 Micro op: $A \leftarrow AB$

b) B op : 00110000
 Micro op: $A \leftarrow A \oplus B$

4-20 What is the register value after an arithmetic shift right? Starting from the initial number 10011100, determine the register value after an arithmetic shift left, and state whether there is an overflow
 right: 11001110 left: 00111000
 no overflow

4-23 What is wrong with the following statements.

- a) $xT: AR \leftarrow AR, AR \leftarrow 0$
- b) $yT: R1 \leftarrow R2, R1 \leftarrow R3$
- c) $zT: PC \leftarrow AR, PC \leftarrow PC + 1$

They all try to assign hardware values to the same register within the same clock cycle

11. given 0000 1111 1010 1010 repeat

a) clear all even bit positions

Bitwise AND
 mask: 1010 1010 1010 1010
 Hex: 0xAAAA

$\therefore$ op AND mask
 is clear even bits

b) set leftmost 4 bits to 1

Bitwise OR
 mask: 1111 0000 0000 0000
 Hex: 0xFF00
 op OR 1111 0000...

c) complement center 8 bits

Bitwise XOR
 mask: 0000 1111 1111 0000
 Hex: 0xFF00

op XOR mask

12. Johnson Counter

a) sequence of states

State 0: 0000 4: 1111
 1: 1000 5: 0111
 2: 1100 6: 0011
 3: 1110 7: 0001

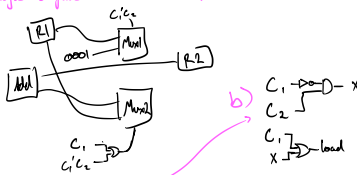
b) How many changes
 on n-bit Johnson counter has
 2n states

13. Given these conditional register transfer

$C1: R1 \leftarrow R1 + R2$ add $R2$ to $R1$

$C1C2: R1 \leftarrow R1 + 1$

a) logic diagram with 4-bit counter & a 4-bit adder



14. Diagram for C3: $R2 \leftarrow R1 + R1 \leftarrow R2$

